4. $2,3,\frac{1}{3}$ 5. 10 6. $6,4,-1$ 7. $\sum \frac{\alpha}{\beta\gamma} = \frac{2ac-b^2}{ad}$

8. $2x^2-3x-20=0$ 11. $x^3+x-12=0$

Exercise 3.2

1. When $k<0$, the polynomial has real roots.
   When $k=0$ or $k=8$, the roots are real and equal.
   When $0<k<8$ the roots are imaginary.
   When $k>8$ the roots are real and distinct.
2. $x^2-4x+7=0$ 3. $x^2-6x+13=0$ 4. x^4-16x^2+4

Exercise 3.3

1. $-3,3,\frac{1}{2}$ 2 . $\frac{2}{3},\frac{4}{3},2$ 3. $\frac{2}{3},2,6$ 4. $k=2,\ 2,\frac{1\pm\sqrt{3}}{2}$

5. $1-2i,1+2i,\sqrt{3},-\sqrt{3},\frac{1+\sqrt{37}}{2},\frac{1-\sqrt{37}}{2}$ 6. (i) $1,\frac{1}{2},3$ (ii) $-1,\frac{1}{2},\frac{3}{4}$ 7. $\pm 3,\pm\sqrt{5}$

Exercise 3.4

1. (i) $\{-2,3,-7,8\}$ (ii) $3,3,3+\sqrt{17},3-\sqrt{17}$ 2. $\left\{1,-2,\frac{-1+2\sqrt{5}}{2},\frac{-1-2\sqrt{5}}{2}\right\}$

Exercise 3.5

1. (i) $n\pi+(-1)^n\frac{\pi}{2}, n\in\mathbb{Z}$, no solution for $\sin x=4$ (ii) $2,-\frac{1}{4},\frac{2}{3}$

2. (i) $x=1$ (ii) no rational roots 3. 4^n 4. $\frac{b^2}{4a},\frac{9a^3}{b^2}$

5. (i) $2,3,\frac{1}{2},\frac{1}{3}$ (ii) $+1,-1,\frac{-3+\sqrt{5}}{2},\frac{-3-\sqrt{5}}{2}$

6. $2,3$ 7. $\frac{1}{3},3,-\frac{1}{2}$ and -2

Exercise 3.6

1. It has at most four positive roots and at most three negative roots.
2. It has at most two positive roots and no negative roots.
4. It has one positive real root and one negative real root.
5. no positive real roots and no negative real roots.

Exercise 3.7

1	2	3	4	5	6	7	8	9	10
(4)	(1)	(3)	(1)	(3)	(4)	(1)	(3)	(1)	(2)

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